Univerza v Mariboru

Fakulteta za elektrotehniko,
računalništvo in informatiko

PREVAJANJE PROGRAMSKIH JEZIKOV

Projektna naloga

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1. UVOD

Cilj projekta je razviti spletno platformo in mobilno aplikacijo, ki pomaga infrastrukturnim inženirjem, da ugotovijo najprimernejša mesta za postavitev (gradnjo) novih reševalnih postaj. Prav tako omogoča izračun in prikaz poti od najbližje trenutne reševalne postaje do lokacije nesreče. S to funkcionalnostjo se optimizira pošiljanje reševalnih/policijskih/gasilskih enot, saj lahko pošljemo na kraj nesreče enoto, ki potrebuje najmanj časa do specificirane lokacije. Torej aplikacija je predvsem namenjena inžinerjem, ne pa vsakdanjim uporabnikom, ki so bili morebiti udeleženi v nesreči.

V tem poročilu predstavljamo analizo in načrtovanje domensko specifičnega jezika (DSL), namenjenega modeliranju geografskih enot, kot so države, regije, mesta, ceste, postaje in nesreče. Jezik omogoča tudi vključevanje dodatnih elementov, kot so parki, hidranti in vizualizacije v obliki okroglih toplotnih map (heatmap). DSL omogoča definiranje strukturiranih in hierarhičnih podatkov, podpira pa tudi osnovne aritmetične in logične operacije.

2. Osnova

Z uporabo našega jezika smo omogočili opis držav, regij, mest, cest, postaje, nesreč, parkov, požarnih hidrantov, heatmap z uporabo geometrijskih oblik. Primer:

```
country "Slovenia" {
 region "Podravska" {
   block ( (15.5,46.6), (15.3, 46.4), (16.0, 46.4), (16.3, 46.5) )
   city "Maribor" {
     street "Koroska cesta" {
       dec_int lanes = 2
       if (2 > 1) {
         dec_string result = "more than one lane"
       } else {
         dec_string result = "one or less lanes"
       }
       dec_double angle = 2.0
       bend ( (15.6,46.5), (15.6,46.5), angle)
       dec_double width = 3.5
       line ( (15.6,46.5), (15.6,46.5) )
     }

     station "Policijska postaja" {
        dec_string addressTemp = "Central Station"
        address = addressTemp
        available_units = 5
        station_type = "Policija"
     }
     accident "Nesreca" {
        dec_string description = "Minor accident with no injuries"
        address = "456 Elm St"
        accident_type = description
     }
     park "Mestni Park" {
         block ( (15.1,46.), (15.1,46.4), (15.8,46.5), (15.6, 46.5) )
     }
     fire_hydrant "Hydrant" {
         point (15.6, 46.5)
     }
     circle_heatmap "Heatmap" {
         dec_string heatmap_name = "Heatmap Example"
         dec_double addRadius = 0.004
         circle ( (15.6,46.4), addRadius )
     }
   }
 }
}
```

2.1 Konstrukti

2.1.1 Enota

- nill

2.1.2 Realna števila

Naš jezik omogoča decimalna in cela števila :

- 12
- 12.3456

2.1.3 Nizi

Nize v našem jeziku predstavljamo :

- “NIZ”

2.1.4 Koordinate

Koordinate so v našem jeziku predstavljene:

- (Expr , Expr)

Pri čemer je Expr lahko spremenljivka, celo število ali decimalno število.

2.1.5 Bloki

- ***Country*** → Državo predstavimo na naslednji način: country "Slovenija" { REGIONS }. Pri čemer ima REGIONS lahko 1 ali več regij.
- ***Region*** → Regijo predstavimo na naslednji način: region “Podravska” { BLOCK CITIES } Pri čemer je BLOCK poligon kateri obdaja, CITIES pa je 1 ali več mest v regiji.
- ***City*** → Mesto predstavimo na naslednji način: city “Maribor” { EXPRESSION }. Pri čemer je EXPRESSION sestavljen obvezno ene ali večih cest, postaj, nesreč in opcijskega dela ki so nič ali več parkov, gasilskih hidrantov in okroglih vročih točk (heatmap).
- ***Street*** → Cesto predstavimo na naslednji način: street “Titova cesta” { OPERATIONS BEND OPERATIONS LINE }. Pri čemer so OPERATIONS deklaracije in if_else stavki, BEND so krivulje in LINE so ravne črte
- ***Station*** → Postajo predstavimo na naslednji način: station “Gasilci Maribor ” { OPERATIONS ADDRESS OPERATIONS AVAILABLE_UNITS OPERATIONS

STATION_TYPE }. Pri čemer so OPERATIONS deklaracije in if_else stavki, ADDRESS je naslov postaje, AVAILABLE_UNITS je celo število, STATION_TYPE je niz.

2.2 Nadgradnja

2.2.1 Elementi

- **Accident** → Nesrečo predstavimo na naslednji način: accident "Prometna nesreča" { OPERATIONS ADDRESS OPERATIONS ACCIDENT_TYPE }, pri čemer OPERATIONS predstavljajo deklaracije in if_else stavki, ADDRESS predstavlja naslov v obliki niza, ACCIDENT_TYPE pa tip nesreče (prav tako niz).
- **Park** → Parko predstavimo z: park “Mestni park” { OPERATIONS BLOCK }, kjer z blokom označimo meje parka, OPERATIONS pa predstavljajo deklaracije in if_else stavki.
- **Fire Hydrant** → Hidrant predstavimo z: fire_hydrant “pozarnihidrant” { OPERATIONS POINT }, kjer s točko (POINT) označimo lokacijo hidranta, OPERATIONS pa predstavljajo deklaracije in if_else stavki.
- **Circle Heatmap** → Vročo točko označimo z: circle_heatmap “vrocatocka” { OPERATIONS CIRCLE }, pri čemer je CIRCLE sestavljen iz koordinate, ki predstavlja središče točke, in polmera (radija) kroga, OPERATIONS pa predstavljajo deklaracije in if_else stavki.

2.2.3 Konstante in spremenljivke

V našem jeziku omogočamo spremenljivke tipa string, int, double, coord. Deklariramo jih na naslednji način:

- dec_int “ime_spremenljivke” = 12
- dec_double “ime_spremenljivke” = 12.3456
- dec_string “ime_spremenljivke” = “tmp”
- dec_coord “ime_spremenljivke” = (12 , 22.3)

2.2.4 Matematični izrazi

V našem jeziku omogočamo osnovni operaciji med spremenljivkami. To sta seštevanje in odštevanje. Primer:

```
bend ( (15.6,46.5), (15.6,46.5), angle + 2.0 )
circle ( (15.66,46.25), 0.004 + 0.002 )
```

2.2.5 Kontrola toka

Naš jezik prav tako omogoča if stavke. If stavek je sestavljen na naslednji način: if (EXPR COMPARE EXPR) {OPERATIONS} else {OPERATIONS} . Pri čemer je else stavek opcijski, EXPR je lahko celo število, decimalno število ali vrednost (var), OPERATIONS predstavljajo deklaracije in if_else stavki, COMPARE pa so naslednji komparatorji:

- == → Preveri enakost
- != → Preveri neenakost
- < → Preveri ali je manjše
- > → Preveri ali je večje

Primer:

```
if (st > 2) {
     dec_string rezultat = "več kot 2"
} else {
     dec_string rezultat = "manj ali enako 2"
}
```

3. FORMALNA DEFINICIJA JEZIKA V BNF

```
Country ::= country string{ Regions }
Regions ::= Region Regions'
Regions' ::= Region Regions' | ε
Region ::= region string { Block Cities }
Cities ::= City Cities'
Cities' ::= City Cities' | ε
City ::= city string { Expression }
Expression ::= Operations MandatoryElementsList AdditionalElementsList
Operations ::= Operation Operations | ε
Operation ::= AssignString | AssignInt | AssignCoord | AssignDouble | IfElse
Expr ::= int Expr' | var Expr' | double Expr'
Expr' ::= + Expr | - Expr | ε
MandatoryElementsList ::= MandatoryElements MandatoryElements'
MandatoryElements ::= Streets Stations Accidents
MandatoryElements' ::= MandatoryElementsList | ε
AdditionalElementsList ::= AdditionalElements AdditionalElements'
AdditionalElements ::= Parks FireHydrants CircleHeatmaps
AdditionalElements' ::= AdditionalElementsList | ε
IfElse ::= if ( Expr Compare Expr ) { Operations } IfElse'
Compare ::= == | != | < | >
IfElse' ::= else { Operations } | ε
Streets ::= Street Streets'
Streets' ::= Street Streets' | ε
Street ::= street string { Operations Bend Operations Line }
Stations ::= Station Stations'
Stations' ::= Station Stations' | ε
Station ::= station string { Operations Address Operations AvailableUnits Operations StationType}
Accidents ::= Accident Accidents'
Accidents' ::= Accident Accidents' | ε
Accident ::= accident string {Operations Address Operations AccidentType}
```

```
Parks ::= Park Parks'

Parks' ::= Park Parks' | ε

Park ::= park string { Operations Block }

FireHydrants ::= FireHydrant FireHydrants'

FireHydrants' ::= FireHydrant FireHydrants' | ε

FireHydrant ::= fire_hydrant string { Operations Point }

CircleHeatmaps ::= CircleHeatmap CircleHeatmaps'

CircleHeatmaps' ::= CircleHeatmap CircleHeatmaps' | ε

CircleHeatmap ::= circle_heatmap string { Operations Circle }

Point ::= point Coordinate

Block ::= block ( Coordinate , Coordinate , Coordinate , Coordinate )

Bend ::= bend ( Coordinate , Coordinate, Angle )

Line ::= line ( Coordinate , Coordinate )

Circle ::= circle ( Coordinate, Expr )

Coordinate ::= ( Expr, Expr )

AssignString ::= dec_string var = AssignString'

AssignString' ::= string | var

AssignInt ::= dec_int var = AssignInt'

AssignInt' ::= int | var

AssignCoord ::= dec_coord var = AssignCoord'

AssignCoord' ::= Coordinate

AssignDouble ::= dec_double var = AssignDouble'

AssignDouble' ::= double | var

Angle ::= Expr

Address ::= address = Address'

Address' ::= string | var

AvailableUnits ::= available_units = AvailableUnits'

AvailableUnits' ::= int | var

StationType ::= station_type = StationType'

StationType' ::= string | var

AccidentType ::= accident_type = AccidentType'

AccidentType' ::= string | var
```

4. TESTNI PRIMERI

```
country "Slovenija" {
   region "Podravska" {
      block((0,0),(0,10),(10,10),(10,0))
      city "Maribor" {
         dec_string spremenljivkaString = "To je spremenljivka"
         dec_coord koordinate = (5, 5)
         dec_int st = 3
         dec_double decimalka = 1.5

         if (st > 2) {
            dec_string rezultat = "več kot 2"
         } else {
            dec_string rezultat = "manj ali enako 2"
         }

         street "Titova cesta" {
            dec_coord a = (0,0)
            dec_coord b = (5,5)
            bend(a, b, 45)
            line(a, b)
         }

         station "Policijska postaja Maribor" {
            address = "Trg svobode 3"
            available_units = 5
            station_type = "policija"
            point(1,1)
         }

         accident "Prometna nesreča" {
            address = "Partizanska cesta"
            accident_type = "prometna"
         }

         park "Park mestni" {
            block((1,1),(1,2),(2,2),(2,1))
         }


         fire_hydrant "Hidrant1" {
            point(2,2)
```

```
            }

            circle_heatmap "ToplotnaCona1" {
                circle((3,3), 10 + st)
            }
        }

        city "Slovenska Bistrica" {
            dec_int spremenljivkaInt = 2
            street "Glavna ulica" {
                dec_coord p1 = (2,2)
                dec_coord p2 = (4,4)
                bend(p1, p2, 90)
                line(p1, p2)
            }
            station "Zdravstveni dom SB" {
                address = "Zdravstvena 12"
                available_units = 3
                station_type = "zdravstvo"
                point(3,3)
            }
            accident "Požar v kuhinji" {
                address = "Industrijska 5"
                accident_type = "požar"
            }
        }

        city "Ptuj" {
            dec_double spremenljivkaDouble = 1.2
            park "Park ob jezeru" {
                block((3,3),(3,4),(4,4),(4,3))
            }

            fire_hydrant "Hidrant2" {
                point(4,4)
            }

            circle_heatmap "Zaznana toplota" {
                circle((5,5), 2.5)
            }
        }
    }
}
```

5. FIRST:

FIRST

FIRST(Country) = {country}

FIRST(Regions) = {region}

FIRST(Regions') = {region,ε}

FIRST(Region) = {region}

FIRST(Cities) = {city}

FIRST(Cities') = {city,ε}

FIRST(City) = {city}

FIRST(Expression)={dec_string,dec_int,dec_coord,dec_double,if,street}

FIRST(Operations)= {dec_string,dec_int,dec_coord,dec_double,if,ε}

FIRST(Operation)= {dec_string,dec_int,dec_coord,dec_double,if}

FIRST(Expr) = {int,var,double}

FIRST(Expr') = {+,-,ε}

FIRST(MandatoryElementsList) = {street}

FIRST(MandatoryElements) = {street}

FIRST(MandatoryElements') = {street,ε}

FIRST(AdditionalElementsList) = {park}

FIRST(AdditionalElements) = {park}

FIRST(AdditionalElements') = {park,ε}

FIRST(IfElse) = {if}

FIRST(IfElse') = {else,ε}

FIRST(Compare) = {==,!=,<,>}

FIRST(Streets) = {street}

FIRST(Streets') = {street,ε}

FIRST(Street) = {street}

FIRST(Stations) = {station}

FIRST(Stations') = {station,ε}

FIRST(Station) = {station}

FIRST(Accidents)= {accident}

FIRST(Accidents')= {accident,ε}

FIRST(Accident) = {accident}

FIRST(Parks) = {park}

FIRST(Parks') = {park,ε}

FIRST(Park) = {park}

FIRST(FireHydrants) = {fire_hydrant}

FIRST(FireHydrants') = {fire_hydrant,ε}

FIRST(FireHydrant) = {fire_hydrant}

FIRST(CircleHeatmaps) = {circle_heatmap}

FIRST(CircleHeatmaps') = {circle_heatmap,ε}

FIRST(CircleHeatmap) = {circle_heatmap}

FIRST(Point) = {point}

FIRST(Block) = {block}

FIRST(Bend) = {bend}

FIRST(Line) = {line}

FIRST(Circle) = {circle}

FIRST(Coordinate) = { (}

FIRST(AssignString) = {dec_string}

FIRST(AssignString') = {string,var}

FIRST(AssignInt) = {dec_int}

FIRST(AssignInt') = {int,var}

FIRST(AssignCoord) = {dec_coord}

FIRST(AssignCoord') = { (}

FIRST(AssignDouble) = {dec_double}

FIRST(AssignDouble') = {double,var}

FIRST(Angle) = {int,var,double}

FIRST(Address) = {address}

FIRST(Address') = {string,var}

FIRST(AvailableUnits) = {available_units}

FIRST(AvailableUnits') = {int,var}

FIRST(StationType) = {station_type}

FIRST(StationType') = {string,var}

FIRST(AccidentType) = {accident_type}

FIRST(AccidentType') = {string,var}

6. FOLLOW:

FOLLOW(Country) = {$}

FOLLOW(Regions) = { } }

FOLLOW(Regions') = { } }

FOLLOW(Region) = {region, } }

FOLLOW(Cities) = { } }

FOLLOW(Cities') = { } }

FOLLOW(City) = {city, } }

FOLLOW(Expression)={ } }

FOLLOW(Operations)= {street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(Operation)= {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(Expr) = {==,!=,<,>,), , ,}

FOLLOW(Expr') = {==,!=,<,>,), , ,}

FOLLOW(MandatoryElementsList) = {park}

FOLLOW(MandatoryElements) = {street,park}

FOLLOW(MandatoryElements') = {park}

FOLLOW(AdditionalElementsList) = { } }

FOLLOW(AdditionalElements) = {park,}}

FOLLOW(AdditionalElements') = { } }

FOLLOW(IfElse) = {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(IfElse') = {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(Compare) = {int,var,double}

FOLLOW(Streets) = {station}

FOLLOW(Streets') = {station}

FOLLOW(Street) = {street,station}

FOLLOW(Stations) = {accident}

FOLLOW(Stations') = {accident}

FOLLOW(Station) = {station,accident}

FOLLOW(Accidents)= {street,park}

FOLLOW(Accidents')= {street,park}

FOLLOW(Accident) = {accident,street,park}

FOLLOW(Parks) = {fire_hydrant}

FOLLOW(Parks') = {fire_hydrant}

FOLLOW(Park) = {park,fire_hydrant}

FOLLOW(FireHydrants) = {circle_heatmap}

FOLLOW(FireHydrants') = {circle_heatmap}

FOLLOW(FireHydrant) = {fire_hydrant,circle_heatmap}

FOLLOW(CircleHeatmaps) = {park,}}

FOLLOW(CircleHeatmaps') = {park,}}

FOLLOW(CircleHeatmap) = {circle_heatmap,park, } }

FOLLOW(Point) = { } }

FOLLOW(Block) = { } }

FOLLOW(Bend) = {dec_string,dec_int,dec_coord,dec_double,if,line}

FOLLOW(Line) = { } }

FOLLOW(Circle) = { } }

FOLLOW(Coordinate) = { }, , ,), dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle }

FOLLOW(AssignString) = {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(AssignString') = {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(AssignInt) = {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(AssignInt') = {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(AssignCoord) = {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(AssignCoord') = {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(AssignDouble) = {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(AssignDouble') = {dec_string,dec_int,dec_coord,dec_double,if,street, }
,bend,line,address,available_units,station_type,accident_type,block,point,circle}

FOLLOW(Angle) = {) }

FOLLOW(Address) = {dec_string,dec_int,dec_coord,dec_double,if,available_units,accident_type}

FOLLOW(Address') = {dec_string,dec_int,dec_coord,dec_double,if,available_units,accident_type}

FOLLOW(AvailableUnits) = {dec_string,dec_int,dec_coord,dec_double,if,station_type}

FOLLOW(AvailableUnits') = {dec_string,dec_int,dec_coord,dec_double,if,station_type}

FOLLOW(StationType) = { } }

FOLLOW(StationType') = { } }

FOLLOW(AccidentType) = { } }

FOLLOW(AccidentType') = { } }

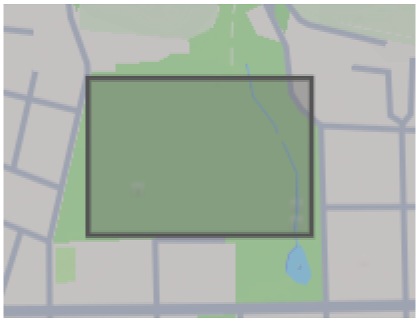
7. GeoJSON

```
           {
             "type": "FeatureCollection",
             "features": [
               {
 "type": "Feature",
 "geometry": {
   "type": "Polygon",
   "coordinates": [[], . . ., [] ]
 },
 "properties": { "name": "Podravska", "type": "region" }
},
{
 "type": "Feature",
 "geometry": {
   "type": "LineString",
   "coordinates": [[], . . . , []]
 },
 "properties": { "name": "Koroska cesta_bend", "type": "bend", "angle":
4.0 }
},
{
 "type": "Feature",
 "geometry": {
   "type": "LineString",
   "coordinates": [[], . . . , []]
 },
 "properties": { "name": "Koroska cesta_line", "type": "line" }
},
{
 "type": "Feature",
 "geometry": null,
 "properties": {
   "name": "Policijska postaja",
   "type": "station",
   "address": "Central Station",
   "available_units": 5,
   "station_type": "Policija"
 }
},
{
 "type": "Feature",
 "geometry": null,
 "properties": {
   "name": "Nesreca",
   "type": "accident",
   "address": "456 Elm St",
   "accident_type": "Minor accident with no injuries"
```

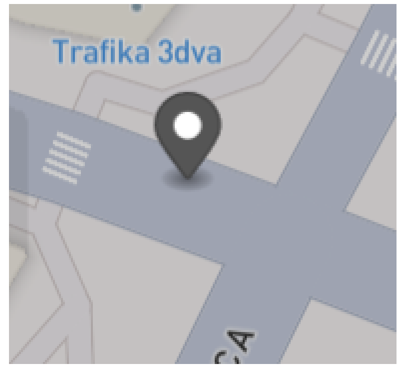
```
 }
},
{
 "type": "Feature",
 "geometry": {
   "type": "Polygon",
   "coordinates": Coordinates
 },
 "properties": { "name": "Mestni Park", "type": "park" }
},
{
 "type": "Feature",
 "geometry": {
   "type": "Point",
   "coordinates": [15.642507100269341,46.55360901515817]
 },
 "properties": { "name": "Hydrant", "type": "fire_hydrant" }
},
{
 "type": "Feature",
 "geometry": {
   "type": "Polygon",
   "Coordinates": [[], . . . , [] ]
 },
 "properties": { "name": "Heatmap", "type": "circle_heatmap", "radius":
1.004 }
}
             ]
           }
```

8. GeoJSON Slike

Park:



Fire Hydrant:



Street:

Circle Heatmap:

Region: